SYSTEM ARCHITECTURE & ENGINEERING WHITEPAPER

FoundryOS

Autonomous AI Marketing & Operations Operating System. Multi-Agent Swarm Orchestration, Semantic Company Brain, Distributed Queues & Enterprise Governance.

DOCUMENT CLASSIFICATION

Technical Whitepaper & Overview

CORE FRAMEWORK

Node.js • TypeScript • PostgreSQL • BullMQ • React

INTELLIGENCE ENGINE

OpenAI GPT-4o / Ollama / pgvector HNSW

DOCUMENT DATE

September 2026 • v0.1.0 Architecture

1. Executive Summary & Vision

FoundryOS is an autonomous AI-driven Marketing & Operations Operating System engineered specifically to empower startups and high-growth companies. Traditional marketing operations suffer from extreme fragmentation: market research is siloed, competitive tracking is manual and quickly obsolete, and campaign strategies diverge from core product truths.

FoundryOS solves this operational chasm by unifying a team of specialized, autonomous AI employees under a centralized, persistent **"Company Brain"** (RAG memory indexed via `pgvector`). It eliminates manual overhead by automating market research, competitor audits, brand positioning, and go-to-market funnel generation within a resilient, distributed, and credit-metered architecture.

Autonomous Agents

SPECIALIZED PERSONAS

Tailored personas for Research, Competitive Intel, Brand Strategy, GTM Funnels, and Multi-Agent Teardowns.

Company Brain

SEMANTIC MEMORY ENGINE

PostgreSQL + pgvector vector embeddings with priority-weighted token budgeting & institutional memory.

Distributed Queues

BULLMQ & REDIS

Fault-tolerant background job execution with automated retries, state recovery, and transactional ledgers.

CORE OPERATING PHILOSOPHY: GROUNDING OVER HALLUCINATION

FoundryOS operates on strict factual grounding: no AI employee generates strategic deliverables without dynamic context retrieval from the Company Brain, verified live tool execution, and deterministic JSON Schema validation.

2. System Architecture & Layered Topology

The system is organized into five decoupled layers engineered for high throughput, memory safety, and model-agnostic flexibility:

LAYER 1: CLIENT APPLICATION & UI

React 18 + Vite SPA

- Live Agent Playground
- Company Brain Studio
- Strategy Pipeline Monitor
- Credit Telemetry Dashboard
- Dark Mode / Modern Aesthetics

▼ HTTP / REST / Server-Sent Events

LAYER 2: API GATEWAY & INTELLIGENT ROUTER

Node.js / Express

- Heuristic Keyword Classifier (<1ms)
- Model-Driven Intent Router
- Zod Request Validation
- Credit Ledger Gatekeeper
- Audit & Error Logging

▼ Dispatches to Specialized Personas

LAYER 3: AUTONOMOUS AI EMPLOYEE SWARM

Dynamic Operating Contracts

- Market Research Analyst
- Competitor Intel Analyst
- Chief Marketing Strategist
- GTM & Funnel Strategist
- FoundryOS Executive Assistant
- Multi-Agent Specialist Swarm

▼ Coordinates Tools & Context Retrieval

LAYER 4: SUBSYSTEMS, TOOLS & EXTENSIBILITY

Modular Engine Services

- Company Brain RAG (pgvector)
- Web Search Plugin (DuckDuckGo / Exa)
- Market Intel (CAC/LTV / Keywords)
- Document & PDF Generation Engine
- Model Context Protocol (MCP) Client
- Flow Engine Pipeline Orchestrator

▼ Inference & Storage Persistence

LAYER 5: FOUNDATION MODELS & PERSISTENCE INFRASTRUCTURE

Multi-Provider & Storage

- OpenAI GPT-4o / GPT-4o-mini (Primary)
- Local Ollama Engine (Fallback)
- PostgreSQL 15+ (pgvector HNSW)
- Redis 6+ (BullMQ Queues & PubSub)

3. Autonomous AI Employees Catalog

FoundryOS provides pre-configured AI employee personas that operate under strict behavioral guidelines, typed input/output schemas (Zod), and domain-specific tool access privileges:

Employee Persona	Pillar / Role	Primary Specialties	Tools	Approval Level
Market Research Analyst MR	Research `marketing_analyst`	TAM modeling, macro industry tailwinds/ headwinds, problem space validation, category trends.	`brain_search` `market_query`	Autonomous (Auto-execute)
Competitor Intelligence Analyst CI	Research `competitor_analyst`	Competitor pricing teardowns, feature matrix audits, temporal shift tracking, differentiation gaps.	`brain_search` `competitor_audit`	Autonomous (Auto-execute)
Chief Marketing Strategist CS	Strategy `strategist`	Company narrative, brand positioning, founder goal alignment, tone & voice governance.	`brain_search`	**Human Approval Required**
GTM & Funnel Strategist FS	Strategy `strategist`	Full-funnel design (TOFU/ MOFU/BOFU), high-conversion offer architecture, campaign KPIs.	`brain_search`	Autonomous (Auto-execute)
FoundryOS Assistant FOS	Strategy `general_assistant`	Executive marketing guidance, startup Q&A, operations navigation, ideation.	`brain_search`	Autonomous (Auto-execute)
Multi-Agent Executive Team	Strategy `multi_agent_team`	360° Collaborative strategy teardown, cross-pillar review, synthesis across all domains.	`brain_search` `web_search` `cac_ltv`	Autonomous (Auto-execute)

Agent Operating Contracts & Schema Validation

Unlike conventional conversational chatbots that produce arbitrary prose, all FoundryOS analytical agents emit strongly-typed JSON outputs conforming to Zod validation schemas:

- `MarketResearchOutputSchema`: Enforces structured market trends, target audience segment opportunities, macro risk assessments, and numerical confidence scores (0.0 to 1.0).
- `CompetitorAnalysisOutputSchema`: Requires detailed competitor teardowns, feature gap matrices, threat levels (`low` | `moderate` | `critical`), and defensive counter-tactics.
- `MarketingStrategyOutputSchema`: Validates positioning statements, value proposition architecture, strategic pillars, and negative brand tone constraints.

- `GtmFunnelOutputSchema`: Outlines structured TOFU, MOFU, and BOFU stages along with offer incentives and KPI benchmarks.

4. The "Company Brain" (Semantic Memory Engine)

The Company Brain is FoundryOS's institutional memory subsystem. It bridges the gap between static company documentation and live agent reasoning using PostgreSQL and the `pgvector` extension.

Vector Indexing (HNSW)

Text embeddings (1536-dimensional via `text-embedding-3-small` or 768-dimensional via Ollama `nomic-embed-text`) are indexed using Hierarchical Navigable Small World (HNSW) graphs, ensuring sub-5ms cosine distance searches across thousands of documents.

Token-Budget Context Builder

The context builder enforces hard token limits (e.g. 4,000 tokens for GPT-4o-mini). It dynamically selects, truncates, and prioritizes knowledge chunks across company profile layers before presenting context to the LLM.

Priority-Weighted Knowledge Layers

When an agent prepares a prompt, the Context Builder populates context according to strict operational priorities:

1. **Core Identity & Product**: Company mission, core value proposition, ICP (Ideal Customer Profile), and product offerings.
2. **Brand Guidelines**: Tone constraints, allowed messaging pillars, and explicit "avoid rules" (prohibited words or angles).
3. **Founder Goals**: Current quarterly milestones, conversion goals, and target metrics.
4. **Dynamic Vector Chunks**: Top-K semantic search matches retrieved from company documentation.
5. **Institutional Memories**: Persistent learnings proposed by agents during previous execution turns.

5. Multi-Provider LLM Engine & Intelligent Router

FoundryOS provides an enterprise-grade inference gateway that ensures zero single-point-of-failure in AI model connectivity.

> **RESILIENT ROUTING STRATEGY**
>
> **Primary:** OpenAI GPT-4o-mini (fast queries, categorization, chat) and GPT-4o (complex multi-step strategy synthesis).
> **Fallback:** Local Ollama instances (e.g., `gemma:2b`, `llama3.2:1b`) automatically engaged if external API limits or network partitions occur.

Two-Tier Agent Router

Incoming user requests pass through a high-efficiency two-tier router to assign the best specialist agent:

- **Tier 1 — Heuristic Classification:** High-speed regex matching evaluates input against high-confidence domain keywords (e.g. "pricing", "competitor", "market size", "TAM", "funnel", "CAC"). Resolves in under 1ms.
- **Tier 2 — Model-Driven Classification:** For nuanced or multi-faceted inquiries, a lightweight classifier evaluates the intent and maps the prompt to the best agent persona, or delegates to the `multi-agent` team.

6. Tool Registry, Plugins & Model Context Protocol (MCP)

Agents in FoundryOS do not hallucinate external facts. They possess direct access to an extensible Tool Registry supporting both native TypeScript plugins and standard MCP (Model Context Protocol) servers.

Plugin Name	Available Tools	Description & Capability
Web Research Plugin `web-search-plugin`	`web_search` `exa_search` `fetch_web_page`	Performs live web queries via DuckDuckGo and Exa AI. Extracts clean markdown from target web pages, enabling agents to study real-time market data.
Market Intelligence `market-intel-plugin`	`keyword_suggester` `cac_ltv_calculator`	Computes Customer Acquisition Cost (CAC), Lifetime Value (LTV), payback period models, and extracts SEO keyword clusters.
Company Brain Plugin `brain-tools-plugin`	`brain_search` `brain_save_insight`	Executes vector similarity search against the Company Brain and archives strategic takeaways into persistent memory.
Document & PDF Plugin `pdf-document-plugin`	`create_pdf_report` `list_pdf_reports` `read_pdf_text`	Generates professional PDF documentation, whitepapers, and audits directly from agent execution results.
Model Context Protocol `mcp.client`	*Dynamic external tools*	Binds external stdio MCP tool servers into the FoundryOS ecosystem with built-in deduplication and memory-leak prevention.

7. Execution Pipelines & Distributed Queue Architecture

Heavy-duty operations—such as comprehensive 360-degree competitor teardowns and automated content calendars—are executed asynchronously using **BullMQ** over **Redis**.

Flow Engine

FoundryOS implements a stateful workflow pipeline that executes sequences of employee tasks. Step outputs are piped to downstream agents as verified context, maintaining transactional state.

Credit Ledger Subsystem

Every LLM invocation and tool call is metered through a transactional credit ledger. Prevents cost runaways by enforcing balance checks prior to workflow initiation.

8. Client Dashboard & User Experience

The FoundryOS user interface is built as a state-of-the-art Single Page Application (SPA) using React, TypeScript, and Vite. Designed with an ultra-sleek, dark-mode cybersecurity aesthetic, it delivers immediate visual clarity into complex multi-agent operations.

Live Agent Playground

Interactive real-time chat with streaming responses, tool invocation badges, and employee persona switching.

Company Brain Studio

Visual management of company profiles, ICP definitions, brand guardrails, document uploads, and vector embedding health.

Strategy Pipeline Monitor

Real-time tracking of BullMQ background jobs, workflow stages, and human-in-the-loop approval requests.

Credit & Health Telemetry

Live monitoring of LLM token burn, credit balance, PostgreSQL vector index status, and Redis connection latency.

9. Core REST API Endpoints

FoundryOS exposes a clean, standardized HTTP REST API for platform integration and automation:

Method	Path	Description
`POST`	`/api/chat`	Executes an agent consultation turn. Automatically routes query, queries Company Brain, and calls required tools.
`GET`	`/api/employees`	Returns full metadata, specialties, and approval rules for all registered AI employee personas.
`POST`	`/api/flows/execute`	Dispatches a multi-stage background flow (e.g. `deep-strategy-flow`) to the BullMQ queue.
`GET`	`/api/flows/:id`	Retrieves the current execution progress, logs, and generated artifacts of a flow.
`POST`	`/api/brain/query`	Executes semantic vector search against the Company Brain knowledge base.
`GET`	`/api/credits/balance`	Returns remaining credit quota and historical consumption breakdown.
`GET`	`/api/health`	System health check verifying database, Redis, and LLM provider connectivity.

10. Deployment & Engineering Specifications

FoundryOS is engineered for enterprise reliability, continuous integration, and rapid deployment.

PRODUCTION READINESS CHECKLIST

- **TypeScript Strict Mode:** Complete zero-warning typing across client, backend, and plugins.
- **Comprehensive Vitest Suite:** 14 test suites covering context budgeting, routing heuristics, queue workers, and error recovery.
- **Memory Isolation:** Circular MCP tool registration guards and bounded process heaps ensure 24/7 uptime.

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